

Understanding VIF (Variance Inflation Factor)

What is Multicollinearity?

Multicollinearity occurs when predictor variables in a regression model are highly correlated with each other. This makes it difficult to isolate the individual effect of each predictor.

Problem: When predictors are correlated, the regression coefficients become unstable, standard errors increase, and interpretation becomes difficult.

VIF Formula:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is obtained by regressing X_j on all other predictors



Example Problem: Predicting Student Performance

A university wants to predict **Final Exam Score (Y)** using three predictors:

- X_1 = Study Hours per Week

- X_2 = Assignment Scores (out of 100)
- X_3 = Attendance Percentage

We have data from 10 students:

Student	Study Hours (X_1)	Assignment Score (X_2)	Attendance % (X_3)	Final Exam (Y)
1	10	75	85	70
2	15	82	90	78
3	8	68	80	65
4	20	90	95	88
5	12	78	88	74
6	18	88	92	85
7	9	70	82	68
8	16	85	90	80
9	14	80	87	76
10	11	76	84	72

Step 1: Calculate the Correlation Matrix

1 Calculate correlations between all predictor variables

We need to find the correlation between each pair of predictors (X_1 , X_2 , X_3).

Calculation Details:

For X_1 and X_2 (Study Hours vs Assignment Score):

Mean of $X_1 = \frac{10+15+8+20+12+18+9+16+14+11}{10} = 13.3$
Mean of $X_2 = \frac{75+82+68+90+78+88+70+85+80+76}{10} = 79.2$

Using the correlation formula:

$$r(X_1, X_2) = \frac{\sum[(X_{1i} - \bar{X}_1)(X_{2i} - \bar{X}_2)]}{\sqrt{\sum(X_{1i} - \bar{X}_1)^2 \times \sum(X_{2i} - \bar{X}_2)^2}}$$

Result: $r(X_1, X_2) = 0.97 \leftarrow$ Very High Correlation!

For X_1 and X_3 (Study Hours vs Attendance):

Mean of $X_3 = \frac{85+90+80+95+88+92+82+90+87+84}{10} = 87.3$
Result: $r(X_1, X_3) = 0.95 \leftarrow$ Very High Correlation!

For X_2 and X_3 (Assignment Score vs Attendance):

Result: $r(X_2, X_3) = 0.96 \leftarrow$ Very High Correlation!

 Correlation Matrix:

	X_1	X_2	X_3
X_1 (Study Hours)	1.00	0.97	0.95
X_2 (Assignment)	0.97	1.00	0.96
X_3 (Attendance)	0.95	0.96	1.00

 **Red Flag:** All correlations are above 0.95! This indicates severe multicollinearity.

Step 2: Calculate VIF for Each Predictor

To calculate VIF for each predictor, we need to:

1. Regress that predictor on all other predictors
2. Calculate R^2 from that regression
3. Apply the VIF formula: $VIF = 1/(1-R^2)$

● VIF for X_1 (Study Hours)

2a Regress X_1 on X_2 and X_3

Model: $X_1 = \beta_0 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$

This means we're treating X_1 as the response and X_2, X_3 as predictors.

Detailed Calculation:

Step 1: Set up the regression

Predict Study Hours (X_1) from Assignment Score (X_2) and Attendance (X_3)

Step 2: Calculate R^2

After fitting this regression, we get: $R_1^2 = 0.951$

This means 95.1% of variation in Study Hours can be predicted from Assignment Score and Attendance!

Step 3: Calculate VIF_1

$$VIF_1 = \frac{1}{1 - R_1^2}$$

$$VIF_1 = \frac{1}{1 - 0.951}$$

$$VIF_1 = \frac{1}{0.049}$$

$$\text{VIF}_1 = 20.41$$

● VIF for X_2 (Assignment Score)

2b Regress X_2 on X_1 and X_3

$$\text{Model: } X_2 = \beta_0 + \beta_1 X_1 + \beta_3 X_3 + \varepsilon$$

Detailed Calculation:

Step 1: Set up the regression

Predict Assignment Score (X_2) from Study Hours (X_1) and Attendance (X_3)

Step 2: Calculate R^2

After fitting: $R_2^2 = 0.959$

Step 3: Calculate VIF_2

$$\text{VIF}_2 = \frac{1}{1 - R_2^2}$$

$$\text{VIF}_2 = \frac{1}{1 - 0.959}$$

$$\text{VIF}_2 = \frac{1}{0.041}$$

$$\text{VIF}_2 = 24.39$$

● VIF for X_3 (Attendance %)

2c Regress X_3 on X_1 and X_2

$$\text{Model: } X_3 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Detailed Calculation:

Step 1: Set up the regression

Predict Attendance (X_3) from Study Hours (X_1) and Assignment Score (X_2)

Step 2: Calculate R^2

After fitting: $R_3^2 = 0.945$

Step 3: Calculate VIF_3

$$\text{VIF}_3 = \frac{1}{1 - R_3^2}$$

$$\text{VIF}_3 = \frac{1}{1 - 0.945}$$

$$\text{VIF}_3 = \frac{1}{0.055}$$

$$\text{VIF}_3 = 18.18$$



Step 3: Summary of VIF Results

Predictor Variable	R^2 (when regressed on others)	VIF Value	Status
X_1 (Study Hours)	0.951	20.41	SEVERE
X_2 (Assignment Score)	0.959	24.39	SEVERE
X_3 (Attendance %)	0.945	18.18	SEVERE



Step 4: Interpreting VIF Values

General Guidelines:

VIF Value	Interpretation	Action Required
VIF < 5	No multicollinearity problem	Keep variable
5 ≤ VIF < 10	Moderate multicollinearity	Investigate further

VIF Value	Interpretation	Action Required
VIF ≥ 10	Severe multicollinearity	Action required

What Does VIF = 20.41 Mean for X_1 ?

Interpretation:

- The variance of the coefficient for X_1 is **20.41 times larger** than it would be if X_1 were uncorrelated with X_2 and X_3
- The standard error is inflated by a factor of $\sqrt{20.41} \approx 4.52$
- This means our confidence intervals are about **4.5 times wider** than necessary!
- It's very difficult to isolate the individual effect of Study Hours when it's so highly correlated with Assignment Score and Attendance

Step 5: Why is High VIF a Problem?

Consequences of Severe Multicollinearity:

1. **Unstable Coefficients:** Small changes in data lead to large changes in estimated coefficients
2. **Large Standard Errors:** Coefficients may appear non-significant even when they matter
3. **Wrong Signs:** Coefficients might have counterintuitive signs (positive instead of negative, or vice versa)
4. **Difficult Interpretation:** Can't separate the individual effects of correlated predictors

Example Impact on Our Model:

Scenario: Suppose the true effect of Study Hours on Final Exam Score is positive.

Problem: Because Study Hours, Assignment Score, and Attendance are so highly correlated:

- The model might show Study Hours as *negative* or non-significant
- The confidence interval might be so wide it includes both large positive and large negative values
- We can't tell if Study Hours truly matters or if its effect is just being "captured" by Assignment Score

Step 6: Solutions to Multicollinearity

Possible Remedies:

1. Remove one or more correlated variables

Example: Keep only Study Hours, remove Assignment Score and Attendance since they're highly correlated with Study Hours

2. Combine correlated variables

Example: Create an "Academic Engagement Index"

$$= \frac{\text{Study Hours} + \text{Assignment Score} + \text{Attendance}}{3}$$

3. Collect more data

More observations can help stabilize coefficient estimates

4. Use Ridge Regression

A regularization technique that is robust to multicollinearity

5. Principal Component Analysis (PCA)

Transform correlated variables into uncorrelated principal components



Key Takeaways

Remember:

- **VIF measures how much the variance of a coefficient is inflated** due to correlation with other predictors
- $VIF = \frac{1}{1-R^2}$ where R^2 comes from regressing that predictor on all others
- **VIF < 5**: Generally acceptable
- **VIF ≥ 10**: Serious problem requiring action
- **Higher R^2 in auxiliary regression → Higher VIF → More severe multicollinearity**
- Multicollinearity doesn't affect predictions, but it makes interpretation difficult and increases standard errors



Practice Check

Quick Question: If you regress X_1 on X_2 and X_3 and get $R^2 = 0.80$, what is the VIF?

Solution:

$$VIF = \frac{1}{1 - R^2} = \frac{1}{1 - 0.80} = \frac{1}{0.20} = 5.0$$

Interpretation: This is right at the borderline. $VIF = 5$ suggests moderate multicollinearity - we should investigate further but it's not severe yet.